

The Local Economic Impact of Data Centers

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Abstract

A hyperscale data center costs over a billion dollars but employs only a few dozen people. We ask whether this investment generates local agglomeration spillovers. Nighttime lights rise sharply at the parcel when construction begins and fade within five kilometers. Advertised salaries and new-firm registrations do not rise, and we rule out increases above 5.5 and 1.6 percent. Hiring and supplier estimates are also near zero but less precise. County data show direct data-processing entry rather than a downstream cluster. Compute firms locate near data centers, but most were already there. Rent estimates are positive but uncertain. Digital capital can transform a site without producing a broad local economic response. The sample covers the cloud-era build-out, not the larger artificial-intelligence campuses that followed.

Keywords: data centers; agglomeration; economic geography; place-based investment.

JEL Codes: R11, R23, O33, H71.

1 Introduction

The physical infrastructure of digital services has become a major component of US capital investment. Data centers consumed roughly four percent of US electricity in 2023, and the rise of artificial intelligence has accelerated new construction [Shehabi et al., 2024, International Energy Agency, 2025]. A hyperscale campus can cost more than a billion dollars and draw hundreds of megawatts of continuous power, yet employs only a few dozen people on site [Alvarez et al., 2026]. At least thirty-five states now offer tax incentives aimed specifically at data centers, with cumulative subsidies approaching twenty billion dollars.¹ Those subsidies are justified in part by the promise of local economic gains, the same promise that motivates place-based incentives more broadly [Slattery and Zidar, 2020, Kline and Moretti, 2014]. This paper asks whether the gains materialize. Our sample covers the cloud-era build-out; the larger artificial-intelligence campuses announced after 2022 are too recent to evaluate here.

The natural benchmark is the canonical result on large plant openings. Greenstone et al. [2010] compare counties that won a large new plant to the runner-up counties that lost it and find that the productivity of incumbent establishments rises in the winning counties. The gain operates through three agglomeration channels: a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. This study provides influential evidence for place-based investment policy. At the same time, subsequent work finds that winner-versus-loser estimates are sensitive to the comparison counties [Patrick, 2016], and that the average large plant produces little evidence of productivity spillovers [Patrick and Partridge, 2022]. The spillover evidence is therefore contested even for manufacturing plants, the historically strongest case for agglomeration.

A data center separates the location of capital from many of the workers, suppliers, and customers that use it. This makes the sector a useful test of the micro-foundations of agglomeration. Labor pooling, input sharing, and knowledge spillovers can all make proximity productive [Duranton and Puga, 2004, Ellison et al., 2010], and it is through those channels that a large employer generates a local multiplier [Moretti, 2010]. A data center engages each of them weakly. It employs dozens of people rather than thousands, buys much of what it needs from regional or national suppliers, and delivers its output over the network to distant customers. Capital scale alone may therefore be insufficient.

The empirical difficulty is that operators screen sites on cheap power, grid capacity, land cost, and fiber, so a host county differs from a non-host in ways that also shape the local economy. We do not find a credible instrument for this selection. The predetermined features that predict siting are correlated with local industrial conditions, while climate and historical human capital do not predict hyperscale locations (Section 3).

We address this with three designs. The within-site design compares an inner ring around each facility with an outer ring around the same site and is our main spatial design for outcomes with a flat pre-period. We date construction from satellite-detected land clearing and restrict the

¹Authors' tally from the Good Jobs First Subsidy Tracker [Good Jobs First, 2024], which records state and local data-center subsidy awards and is one of the sources for our facility registry (Section 2).

capital analysis to sites without an earlier data center within two kilometers. We also assemble 84 announced hyperscale-sized projects that were not built, of which 52 have satellite coverage. These projects crossed an observable announcement and siting threshold, but they are not literal runner-up locations. They are newer, farther from existing data-center clusters, and less likely to have a named hyperscale sponsor than the built sites. We therefore use them as an announced near-miss benchmark, not as an assumption that the groups are balanced. County panels provide a third diagnostic for industry outcomes, although suppression and pre-opening trends limit what these data establish.

We first establish that the capital footprint is large and local. At isolated sites, nighttime lights rise by 38 percent within the first kilometer when construction begins and decay to near zero by five kilometers [Henderson et al., 2012]. The near-miss comparison agrees on timing and, once the two are placed on a common scale, on magnitude. Daytime imagery confirms that vegetation clears when the lights appear. The size of the response varies across construction cohorts, and both designs show the same gradient, which tracks how dark the parcel was to begin with.

The evidence for agglomeration depends on the outcome. Advertised salaries and new-firm registrations do not rise, and their confidence intervals exclude increases above 5.5 and 1.6 percent. A national county analysis of business applications gives a similar result. Hiring is bounded more loosely, at 17 percent, and supplier postings loosely enough to be uninformative. Workplace employment rises before opening rather than breaking at the event. At the county level, data-processing establishments rise by 27 percent, consistent with the direct entry of the facility. Apparent establishment growth in broader information and professional services is also present before opening.

These bounds require a control area the facility does not reach. Because hyperscale sites cluster, another data center sits inside the conventional 25 to 50 kilometer control ring at 88 percent of our built sites, which would push any labor-market response toward zero mechanically. We therefore compare the 0 to 25 kilometer zone with a 50 to 150 kilometer control. The nulls survive and tighten.

The geography of compute reinforces this distinction. Compute-intensive firms locate closer to data centers than other firms, but 69 percent of those within 25 kilometers predate the nearest facility. The median metropolitan data center has thousands of compute firms nearby, while the median exurban data center has 41. The two populations share metropolitan infrastructure, skilled labor, and access to the power and fiber backbone. Their geographic overlap does not imply that the data center created the cluster.

The non-labor evidence is more suggestive. Synthetic-control rent estimates are two to five percent, but the announced near-miss estimate is imprecise and the synthetic comparison does not support formal placebo inference. Multifamily permitting and net migration do not rise. School property-tax revenue and county highway spending have positive point estimates, but neither survives adjustment across the fiscal outcomes. Foot traffic and commercial spending show no robust change. Residential electricity prices, reported in the appendix, do not rise for the cloud-era cohorts

in our sample.

Concurrent county-level studies find positive effects of data-center entry or capacity on employment, establishments, income, construction, and house prices [Bahar and Wright, 2026, Alvarez et al., 2026, Yue and Zeng, 2026]. Our question is different. County estimates combine the direct facility, construction, and growth already under way in selected locations. We ask whether construction or opening produces a discrete response among nearby workers, suppliers, and firms, and we trace where compute users were located first. Section 7 works through the comparison, including a component of our own earlier county evidence that finer data revise. The results reconcile a large direct investment with limited evidence of a new local cluster. This distinction matters as place-based investment shifts from labor-intensive plants toward labor-light digital capital.

2 Data

Data center registry. We build a master registry of US data centers from several sources. The starting point is the Open Source Data Center Atlas (Pacific Northwest National Laboratory’s IM3 project), which identifies 1,242 facilities from OpenStreetMap footprints tagged as data centers, with coordinates and, where available, square footage and operator. We add operational-date information from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, and trade press. We also conduct targeted searches of state announcements, specialty wholesale operators, AI-wave operators, and real-estate investment trust filings. The final registry contains 1,494 facilities. The geocoded analysis sample contains 1,118 facilities, including 341 classified as hyperscale. Of the hyperscale facilities, 315 have high- or medium-confidence analysis dates inside the 2012–2024 VIIRS window. We classify facilities as hyperscale based on operator identity (Amazon, Microsoft, Google, Meta, Apple, Oracle, plus hand-curated large specialty and AI-wave campuses). We do not impose a megawatt threshold, since capacity is not consistently observed. We also assemble an announced near-miss cohort for the built-versus-near-miss benchmark. It contains 84 announced projects that were not built, of which 52 have the coordinates and satellite coverage to enter the nighttime-lights design, and 49 of those are away from operating campuses and enter the isolated capital sample. Section 3 reports the cohort’s sponsor composition and shows that the capital result is robust to restricting the control group to projects with a named hyperscale tenant.

Table 1: Sample construction

Sample restriction	N facilities
Master registry (all sources, all years)	1,494
geocoded and dated analysis backbone	1,118
with an analysis date in 2012–2024	1,110
with clean-confidence analysis date	1,088
Hyperscale-tier facilities	341
with an analysis date in 2012–2024	334
with clean-confidence analysis date	315
with satellite-detected construction date	87
Isolated built sites in capital design	43
Isolated announced near-miss sites	49

Notes: Sample is constructed from a master registry of 1,494 US data centers compiled from the IM3 OpenStreetMap base, operator press releases, state economic-development announcements, Good Jobs First subsidy tracker entries, and industry press. Hyperscale facilities include the six major cloud operators and hand-classified large specialty and artificial-intelligence campuses. The date audit (`cf_395`) stores announcement, groundbreaking, construction, planned-completion, and operational events separately. Clean dates are high or medium confidence; mechanically projected openings, planned-completion dates without verified operation, and unresolved source conflicts are low confidence. The capital design further requires a satellite-detected construction date and no earlier data center within two kilometers.

Nighttime lights. We extract monthly mean radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band Monthly Composite for April 2012 through December 2025, in concentric rings of 0–1, 1–2, 2–5, 5–10, 10–25, and 25–50 kilometers around each facility, aggregated to annual means. We classify each facility as isolated if no other data center within two kilometers opened earlier, and use the isolated, construction-dated sample for the headline capital-footprint estimate.

Local labor, housing, and spending panels. We draw four panels from Dewey Data. LinkUp Job Records [LinkUp, 2025] measures job postings by employer, geocoded location, ring, and quarter. RentHub Rental Data [RentHub, 2022] measures monthly median list prices by ring. We measure consumer spending with SafeGraph Spend Patterns [SafeGraph, 2022] and foot traffic with Advan Weekly Patterns [Advan Research, 2025]. The rent, spending, and foot-traffic panels enter the within-site ring design dated to operations, with state-by-month fixed effects. The posting and salary panels are extracted out to 150 kilometers, so that the labor outcomes can use a control zone beyond the reach of the facility’s own cluster (Section 3.3).

County employment. For the county-level analysis, we use industry employment and establishment counts from the Quarterly Census of Employment and Wages. We retain the Bureau of Labor Statistics disclosure flag, which is essential because the public files record suppressed employment as zero even though the value is not observed. We focus on data processing (NAICS 518),

telecommunications (517), web and information services (519), IT consulting (5415), and professional, scientific, and technical services (541). We end the 519 series in 2021 because the 2022 NAICS revision moved major components of that sector to 513 and 516. Section 5.5 reports both the disclosure rates and the pre-trend diagnostics that limit causal interpretation of these county outcomes.

Instrument-search inputs. The instrument search draws on predetermined county features, namely 345 kV transmission-substation density, the InterTubes long-haul fiber network [Durairajan et al., 2015], the 1980 county share of urban college population (NHGIS), station-level climate normals (CustomWeather), and the EIA-860 generation fleet (Appendix B).

2.1 Dating construction and operations

We preserve announcement, groundbreaking, construction, planned-completion, and operational dates as separate events. We assign high confidence to operational dates reported by operators or public agencies. We assign medium confidence to less detailed sources when the year agrees with an independent registry or satellite evidence. Announcement-based projections, planned-completion dates, and unresolved source conflicts remain low confidence and are excluded from analyses restricted to reliable dates. This classification leaves 315 high- or medium-confidence hyperscale dates, 19 low-confidence dates, and 7 facilities dated outside the 2012–2024 window. For a smaller sample, we recover construction timing from daytime satellite imagery. A hyperscale build begins with land clearing. In Landsat and Sentinel-2 imagery, clearing appears as a sustained drop in the parcel’s vegetation index combined with a rise in its built-up-surface index. Vegetation cover at the parcel is flat until the construction year and falls sharply afterward. As an independent check, we match Federal Aviation Administration obstruction-evaluation filings to our projects. The imagery sample contains 43 hyperscale facilities, while the independent FAA matches contain 64 facilities within 500 meters and 119 within one kilometer (Section 3.6). These independent dates validate the satellite construction timing used in the capital design.

3 Research Design

Data centers do not locate at random. Operators screen on cheap power, available grid capacity, large parcels of low-cost land, fiber proximity, and tax policy, so a county that hosts a data center differs from one that does not in ways that also move the local economy. A cross-county comparison would confound the data center with the conditions that attracted it. We document this selection directly. We test a range of predetermined siting predictors as potential instruments and reject each for reasons reported in Appendix B. The predictors that strongly predict siting are power, industrial land, and fiber, and each is itself correlated with local industrial conditions. Climate and historical human capital are less likely to violate the exclusion restriction, but they do not predict hyperscale siting.

We therefore use three designs that answer different parts of the question. The within-site ring comparison provides the main spatial estimate for outcomes with a flat pre-period. It compares change close to a facility with change farther from the same site, absorbing time-invariant siting differences. Announced projects that were not built provide an external near-miss benchmark. Unlike the runner-up counties in [Greenstone et al. \[2010\]](#), these projects are not known finalists in the same site-selection process, so we do not assume that they are balanced with built sites. County panels provide an aggregate industry diagnostic, but suppression and pre-opening trends prevent us from treating most of these estimates as clean downstream employment effects.

3.1 Announced near-miss facilities

An announced and sited facility that was never built provides a benchmark for a built facility. Both groups crossed an observable siting threshold, but the comparison does not eliminate selection between projects that proceed and projects that do not. We assemble 84 announced-then-cancelled hyperscale-sized projects from trade-press coverage, regulatory filings, and operator announcements, of which 52 enter the nighttime-lights design, and estimate, for each outcome,

$$y_{i,r,t} = \sum_k [\beta_k \mathbf{1}[t - t_i^c = k] \times \text{Near}_{i,r} \times \text{Built}_i + \delta_k \mathbf{1}[t - t_i^c = k] \times \text{Near}_{i,r}] + \alpha_{i,r} + \lambda_t + u_{i,r,t}, \quad (1)$$

where Built_i marks facilities actually built, t_i^c is the satellite-detected construction-start date for built sites and the announcement date for cancelled sites, $\alpha_{i,r}$ is a facility-by-ring fixed effect, and λ_t is a calendar-year fixed effect. For the capital footprint, $y_{i,r,t}$ is the log of mean radiance. The cancelled projects do not have an observed or consistently reported scheduled start, so their timing is not symmetric with the built group. Announcement typically precedes construction by a year or more, which places part of the built group’s own announcement activity inside its pre-period and, if anything, understates the jump. The path of β_k traces the change around built facilities relative to the change around cancelled sites. For the capital footprint, the comparison is especially sharp because a parcel that is never built does not light up, whatever the reason for cancellation.

The announced near misses vary in sponsor commitment. Of the 84 candidate projects we track, 14 had a named hyperscale operator publicly attached at cancellation, 19 were sponsored by established data-center operators without a named tenant, and the remaining 51, about 60 percent, were speculative developments by real-estate sponsors. Speculative developments have no signed tenant or confirmed grid allocation and are less informative about the counterfactual for a committed project. For the capital footprint, however, a parcel that is never built does not light up whatever the sponsor. Restricting the controls to the nine isolated projects with a named hyperscale operator leaves the groundbreaking-year jump at 72 percent, against 70 percent in the full isolated sample, although that subset carries a somewhat less flat pre-period and is too small to support estimates at longer horizons. The full cancelled sample is also limited, and most cancellations followed local opposition, so cancelled sites may sit in somewhat less development-friendly places.²

²Because most near-miss projects were announced late in the sample, the near-miss group contributes few post-

This is in part why we treat this design as a benchmark and pair it with the ring comparison below. In addition, built sites are dated by satellite-detected land clearing, while near-miss sites have less reliable scheduled dates from filings and trade press. The independent regulatory-filing dates described in Section 3.6 validate the built-site dates.

Table 2 makes the limits of the comparison explicit for the exact capital-footprint sample. Built and near-miss sites have similar local population and high-voltage access. They differ substantially in event year, baseline radiance, proximity to an earlier data center, region, and sponsor commitment. Facility-by-ring fixed effects absorb the level differences, and the pre-event path tests whether the inner-to-outer change was already diverging. The near-miss group is also concentrated in recent years, with one to five sites per calendar year before 2020 against 21 to 49 from 2020 onward, while 26 of the 43 built sites break ground in 2017 or 2018. Section 4 therefore reports the estimate separately by construction cohort and shows that the ring design, which uses no near-miss control, reproduces the same cohort pattern. The table nevertheless rules out interpreting the near misses as conventionally balanced runner-up sites.

event observations. Coverage at event years zero through four is 48, 21, 5, 3, and 1 sites, so the built-versus-near-miss path at longer horizons is identified primarily from changes among built sites relative to the common pre-event trend rather than from post-period movement among near-miss sites. Section 4 quotes magnitudes only through the year after groundbreaking for this reason.

Table 2: Built and announced near-miss sites in the capital-footprint sample

Characteristic	Built mean	Near miss mean	Standardized difference
Event year	2018.81	2024.37	-4.21
Log population within 10 km	10.54	10.42	+0.06
Pre-event log radiance, 0–1 km	0.54	2.26	-1.28
Log distance to earlier data center	2.35	3.64	-1.19
Log distance to ≥ 230 kV substation	1.87	1.90	-0.04
Log distance to long-haul fiber	2.15	1.87	+0.29
Share Northeast	0.00	0.06	-0.36
Share Midwest	0.23	0.41	-0.38
Share South	0.35	0.43	-0.16
Share West	0.42	0.10	+0.76
Share with named hyperscale sponsor	1.00	0.18	+2.95

Notes: Exact isolated-site sample used at event year zero in Figure 1, namely 43 built sites and 49 announced near-miss sites. Built sites are dated by satellite-detected construction start; near-miss sites are dated by announcement. The standardized difference is the difference in group means divided by the square root of the average of the two group variances, so a value of one means the groups differ by one pooled standard deviation. Distances are in kilometers before applying $\log(1 + x)$. Population is based on 2020 Census tract population centroids. High-voltage access uses substations rated at least 230 kV; long-haul fiber uses the InterTubes conduit map. The table is a diagnostic, not a claim that the groups are balanced.

3.2 Within-site rings

The near-miss comparison provides an external benchmark but is too small and imbalanced to carry every outcome. Our main spatial design compares an inner ring around a facility to the same facility’s 25 to 50 kilometer outer ring,

$$\log(1 + \text{NTL}_{i,r,t}) = \sum_k \beta_k \mathbf{1}[t - t_i^c = k] \times \text{Near}_{i,r} + \alpha_{i,r} + \gamma_t + \varepsilon_{i,r,t}, \quad (2)$$

where i indexes the facility, r the ring, t the year, t_i^c the facility’s construction-start year, $\text{Near}_{i,r}$ marks the inner ring, $\alpha_{i,r}$ is a facility-by-ring fixed effect, and γ_t is a calendar-year fixed effect. The facility-by-ring fixed effect absorbs any time-invariant difference between a site’s inner and outer ring, including the operator’s siting preferences, the local industrial structure, and grid and fiber access. Identification comes from the change in the inner ring relative to the outer ring of the same site.³

³With two rings, replacing γ_t with a ring-by-year term would absorb the average inner-versus-outer gap in each calendar year and leave the event-study coefficients identified only from differences in construction timing across sites. We use the calendar-year specification throughout. Section 4 reports the estimate under the log of radiance as well, so that its magnitude can be compared with the near-miss design.

Two features of the design address a pre-trend present in a simpler version of this specification. First, we date the event to construction start, not to the operational opening, to avoid confusing pre-period construction activity with operations (Section 2.1). Dated to construction, the inner-ring brightness is flat before groundbreaking and jumps at the groundbreaking date.

Second, we restrict the headline estimate to isolated sites, meaning facilities with no earlier data center within two kilometers. A new facility built beside an existing one has a brighter inner ring even before construction begins, which could generate a false pre-trend. The isolated, construction-dated event study has a flat pre-trend and then a jump of 38 percent in the construction year that builds to a larger footprint over the following years. Fitting a separate linear pre-trend to each site’s inner-minus-outer path over the four pre-construction years produces an observed-minus-predicted construction-year change of 38 percent, with a campus-bootstrap 95 percent interval of 15 to 71 percent.

We report ring estimates only for outcomes whose pre-construction estimates are flat. The capital footprint, rents, advertised wages, and firm entry pass this test. Job-posting counts and workplace employment do not, so for those outcomes we rely on the comparisons described next and do not interpret the inner-versus -outer ring estimates in either direction. To make these choices transparent, Table 4 reports every outcome under every design in which it can be estimated, with the estimate, its precision, and the reason any design is excluded for that outcome.

3.3 A control ring that is itself treated

The within-site design assumes that the 25 to 50 kilometer outer ring is untreated. That assumption is strained. Hyperscale facilities cluster, and another data center sits inside the outer ring at 88 percent of the built sites in the capital sample. At 81 percent of them it had already opened before the focal site broke ground, and the median site has ten of them. If labor-market and supplier effects operate at commuting-zone scale, which is roughly the scale of the outer ring itself, then the control ring absorbs part of the response and the within-site estimate is biased toward zero.

This matters asymmetrically. It works against the capital result, which survives anyway. But it works in favor of our null results, which are the substance of the paper, so we cannot let those rest on the within-site comparison alone. For the labor outcomes we therefore replace the outer ring with a far control, pooling the 0 to 25 kilometer zone and comparing it with a 50 to 150 kilometer control around the same facility. Table 3 reports both.

The nulls survive, and they tighten. Advertised salary moves by +1.3 percent against the far control, with a 95 percent interval of $[-2.7, +5.5]$, compared with -0.8 percent and $[-6.2, +4.8]$ within site. Total postings move by +5.2 percent, with an interval of $[-5.3, +16.8]$, against -4.6 percent and $[-17.5, +10.3]$ within site. The far control is the more informative comparison in both cases, and we use it for the labor outcomes in Section 5. The exercise also flags one within-site result that does not survive. Colocation postings rise by 14.6 percent at 0 to 5 kilometers against the outer ring ($p = 0.03$), but by 4.1 percent against the far control ($p = 0.52$).

Table 3: Labour outcomes under two control zones

Outcome	0–5 vs 25–50 km		0–25 vs 50–150 km	
	Estimate	95% CI	Estimate	95% CI
<i>Panel A: Hyperscale</i>				
Total postings	-4.6%	[-17.5, +10.3]	+5.2%	[-5.3, +16.8]
Advertised salary	-0.8%	[-6.2, +4.8]	+1.3%	[-2.7, +5.5]
<i>Panel B: Colocation</i>				
Total postings	+14.6%	[+1.8, +29.0]	+4.1%	[-7.7, +17.4]
Advertised salary	+3.2%	[-1.0, +7.7]	+0.6%	[-3.8, +5.1]

Notes: Trend-break estimates at the operational opening, monthly panel, 24 months either side, standard errors clustered on the facility. The left pair uses the paper’s within-site control ring; the right pair replaces it with a 50–150 km control. Another data center sits inside the 25–50 km control ring at 88 percent of the built sites in the capital sample, and had already opened before the focal site broke ground at 81 percent; the median site has 10 of them. A control ring that is itself treated biases the within-site estimate toward zero.

3.4 Demand-side outcomes

We use the same within-site ring comparison in equation (2) for three demand-side outcomes, namely residential rents, consumer spending, and foot traffic. We date these outcomes to the operational opening rather than to the construction start. The rent, spending, and foot-traffic panels are monthly and span 2020 to 2023, so we replace the calendar-year fixed effect with a state-by-month fixed effect, which absorbs the differential work-from-home and urban-recovery shocks across metropolitan areas in those years. The announced near-miss design provides a second comparison where those data are available. Relative to near-miss sites, rents rise modestly at built sites, foot traffic shows no robust change, and commercial spending does not change in either comparison. The rent, spending, and foot-traffic estimates enter the incidence analysis in Section 6.

3.5 Employment and county-industry diagnostics

Local labor demand is harder to identify than the capital footprint, so we measure it in two ways. The first approach counts openings advertised near each facility, using the far control described above.

The second approach is at the county level, because a data center’s own employees and any firms that cluster around it need not fall inside a single ring. We use Quarterly Census of Employment and Wages employment and establishment counts by industry to estimate

$$\log(1 + \text{Emp}_{c,t}^s) = \sum_k \theta_k^s \mathbf{1}[t - t_c^o = k] \times \text{DC}_c + \mu_c + \lambda_t + u_{c,t}, \quad (3)$$

for industry s , county c , and the county’s first data-center year t_c^o , with county and year fixed effects. We also estimate cohort-by-event-time effects using not-yet-treated counties as controls

and state-cluster bootstrap inference, following [Callaway and Sant’Anna \[2021\]](#). These diagnostics limit how we use the county data. Between 27 and 68 percent of the detailed employment cells are suppressed, depending on sector, and establishment counts in several sectors rise before opening. We therefore use these outcomes to show why the county evidence cannot cleanly identify either a downstream gain or a precise zero. Section 5.5 reports the diagnostics and results in full.

3.6 Credibility of the capital-footprint design

The capital-footprint estimate is robust to the usual checks on a ring design. Dropping each facility in turn leaves the estimate in a narrow band, so no single site drives it. A specification that omits the campus itself and uses the 1 to 2 kilometer ring as the inner comparison returns the same spatial decay. Because most hyperscale sites sit near one another, we cluster standard errors at the campus, rather than the facility, level, which widens the interval but leaves the effect statistically significant. Most important for construction timing, we re-date the facilities from Federal Aviation Administration (FAA) obstruction-evaluation filings. These regulatory timestamps are recorded years before any radiance response, and we match them to facilities by location without using satellite data. The construction-dated capital footprint is similar in sign and total magnitude under these independent dates. The FAA filing typically precedes physical construction, so this is a timing validation rather than an exact alternative groundbreaking date.

Because both the ring and near-miss designs use staggered construction dates, we also report a cohort-robust estimator that compares each construction cohort only with never-built sites observed in the same calendar years (Section 4). That estimate is imprecise and fails its lead test, so we report it as an inconclusive diagnostic rather than as corroboration.

In Section 6 we conduct one additional check for the rent result. Rents rose faster near metropolitan cores than in their surrounding areas over 2020 to 2023, and a data center’s inner and outer rings sit at different points on that gradient, so the within-site comparison can pick up the gradient rather than the facility. We therefore build a synthetic control for each facility from population-matched locations with no data center. This comparison accounts for common growth and isolates the data-center-specific rise.

Table 4 records which design carries the interpretation for each outcome. This routing matters because no single comparison is credible for all outcomes.

Table 4: Outcome estimates across research designs

Outcome (hyperscale)	Built vs. near miss	Within site or far control	County evidence
Capital (nighttime lights)	+70% ($p < 0.01$)	+38% ($p < 0.01$) ^a	— ^b
Residential rent	+4.3% ($p = 0.23$)	+2 to +5% ^c	— ^b
Advertised salary	−11.6% ($p = 0.54$)	+1.3% ($p = 0.53$) ^d	— ^b
Job postings	+4.0% ($p = 0.75$)	+5.2% ($p = 0.35$) ^d	— ^b
Supplier postings	— ^e	+5.9% ($p = 0.61$) ^d	— ^b
Foot traffic	−12.0% ($p = 0.12$) ^f	excluded ^g	— ^b
Consumer spending	+4.6% ($p = 0.60$)	+8.5% ($p = 0.50$) ^h	— ^b
Firm entry (SoS registrations)	— ^e	−0.5% ($p = 0.62$) ⁱ	+1.4% ($p = 0.49$) ^j
Workplace employment	— ^e	excluded ^g	inconclusive ^k

Notes: Post-event estimates for hyperscale facilities at the innermost ring each design supports. Bold marks the single design used to interpret each outcome. The built-versus-near-miss capital estimate is the isolated-sample construction-year jump; the two groups are not balanced on several observable characteristics (Table 2), and magnitudes at longer horizons rest on few near-miss controls (Section 4).

^a Agrees with the near-miss estimate once both use the same transform; the gap is $\log(1+x)$ against $\log(x)$, not the comparison group (Table 5).

^b Not estimated at the county level; no county analog of the ring outcome.

^c Synthetic-control headline; the within-site estimates (+6.6 to +9.4%) are an upper bound (Section 6).

^d Pooled 0–25 km zone against a 50–150 km control, because another data center sits inside the 25–50 km ring at 88 percent of built sites (Section 3.3). The within-site figures are −0.8% for salary and −4.6% for postings.

^e No panel around near-miss sites exists for this outcome (business registrations and LODES cannot be reconstructed around never-built sites at ring scale).

^f Attenuates to −3.0% ($p = 0.68$) on the balanced panel, so we do not interpret it as reliable evidence of displacement.

^g Excluded because the pre-opening coefficients show an inner-versus-outer trend that the ring design cannot separate from an effect (Section 3).

^h Reported with caveats in Section 6; passes a random-donor comparison.

ⁱ Hyperscale-only sample using updated opening dates; the pooled all-archetype estimate is +0.7% ($p = 0.24$).

^j Census Business Formation Statistics, national county design.

^k Data-processing establishments rise with direct entry. Detailed QCEW employment is heavily suppressed, while establishment counts in several downstream sectors rise before opening. Estimates that preserve the suppression flags and use not-yet-treated controls therefore do not identify either a downstream gain or a precise zero (Section 5.5).

4 The capital footprint

4.1 Within-site ring design

Our main spatial estimate compares the inner ring around a facility with the same facility’s 25 to 50 kilometer outer ring, dated to the construction start recovered from satellite-detected land clearing (Section 2.1). Two features of the design matter.

The first is dating. A hyperscale campus takes two to three years to build, so dating the event to the operational opening places the construction phase, when the parcel is physically changing, before the event window and reads it as a pre-trend. Observing the construction start and the operational opening separately prevents the construction period from being read as an opening-date pre-trend.

The second is the sample. Roughly half of the hyperscale sites with a satellite-detected construction date sit within two kilometers of an earlier data center, so the site inherits its neighbor’s brightness in its inner ring before it breaks ground. Splitting the sample makes this explicit. At sites with an earlier neighbor within two kilometers, inner-ring brightness rises about 50 percent in the years before construction. At isolated sites, where the inner ring is just the future parcel, the pre-construction excess is 7 percent and the event study is flat before groundbreaking. A joint test of the pre-construction coefficients does not reject ($p = 0.37$), and the jump is 38 percent in the construction year. Fitting a separate linear pre-trend to each site’s inner-minus-outer path over event years -4 through -1 and extrapolating it to event year zero returns a 38 percent jump, with a campus-bootstrap 95 percent interval of 15 to 71 percent.

4.2 Announced near-miss benchmark

Figure 1 reports the announced near-miss benchmark from equation (1), which compares the inner-ring (0 to 1 kilometer) path at built hyperscale sites with announced projects that were not built. We focus on *isolated* sites, the 43 built facilities with no earlier data center within two kilometers of the parcel and the 49 near-miss sites away from operating campuses. As Table 2 shows, these groups are not balanced on several observable characteristics.

The pre-construction coefficients are between 8 and 11 percent below the omitted year. They are jointly indistinguishable from zero when clustered by campus ($p = 0.24$), and their linear slope is $+3.5$ log points per year ($p = 0.22$). The event-study estimate is 70 percent in the construction year and reaches roughly 231 percent above the near-miss benchmark a year later. Clustering standard errors at the campus level widens the intervals by roughly 60 percent and leaves every post-groundbreaking coefficient statistically significant. The measured footprint reflects activity at the parcel, its security and parking lighting, substations and cooling equipment, and continuous operation.

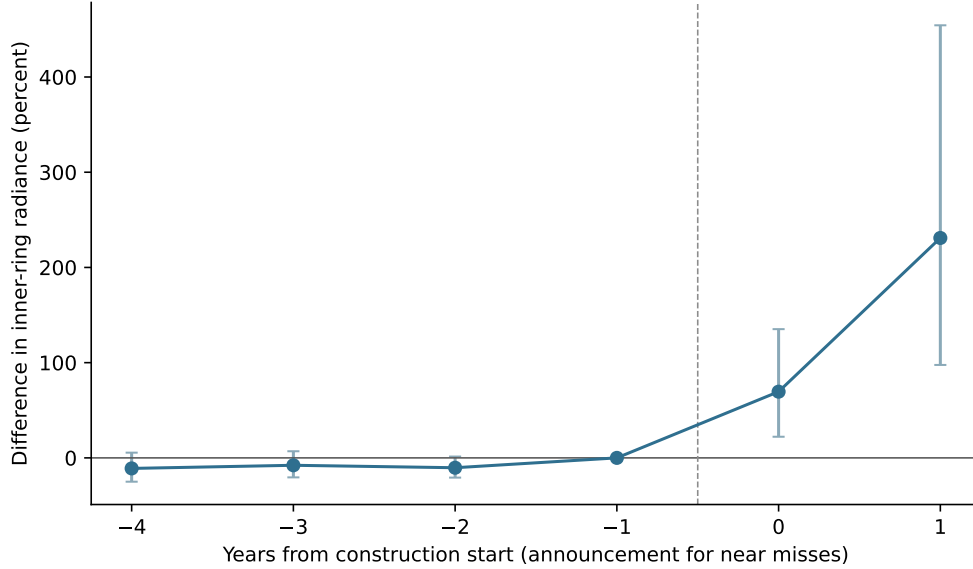


Figure 1: Capital footprint at built versus announced near-miss sites

Alt text: The event-study line is flat for four years before construction, rises sharply in the construction year, and rises further one year later. Confidence intervals widen after construction.

Notes: Event study of equation (1): the inner-ring (0 to 1 km) log nighttime-lights path at 43 isolated built hyperscale facilities relative to 49 isolated announced near-miss facilities, omitting the year before. Built facilities are dated by satellite-detected land clearing; near-miss facilities are dated by announcement year. Standard errors are clustered on two-kilometer campus groups. Near-miss coverage is 48 sites in the construction year and 21 one year later. Table 2 reports observable differences between the groups, and Table 5 reports the estimate across designs, transforms, and construction cohorts.

Two properties of this benchmark govern how it should be read. The first is coverage at longer horizons. The near-miss cohort is concentrated in recent announcement years, so 48 control sites identify the construction-year contrast and 21 the year-one contrast, but five or fewer identify the contrasts beyond that. We end the figure after year one and do not use the later coefficients.

The second is that the footprint belongs to the campus rather than to a single building, and in two distinct ways. Thirty-two of the 43 isolated built sites add a sibling building within two kilometers during the event window, which is why the full-sample path keeps rising through year four. Separately, the 11 sites that stay single-building throughout are much smaller projects, and their construction-year jump is 7.7 percent with a standard error of 6.9 percent. Sibling phasing cannot account for that gap, because no sibling has been added at the construction year itself. The two groups differ in project scale. The estimate is therefore the response of a large multi-building campus rather than of one building, and we report it that way because the campus is what a jurisdiction recruits.

4.3 Reconciling the two designs

The two designs agree on timing and shape, and they agree on magnitude once they are placed on the same scale. Table 5 reports both. The apparent gap between 70 and 38 percent is the outcome

variable rather than the comparison group. The near-miss design takes the log of mean radiance, while the ring design uses $\log(1 + \text{NTL})$, which compresses changes at the low pre-construction radiance of these parcels. Estimated on the log of radiance, the ring design returns 63 percent in the construction year and 226 percent a year later, against 70 and 231 percent for the near-miss design. A Poisson specification on radiance levels returns 59 percent. Three comparisons that differ in control group and functional form therefore land in the same range.

The size of the footprint varies systematically across construction cohorts, and both designs show the same gradient. Sites that broke ground in 2018 or earlier show a construction-year change of 97 percent in the near-miss design and 95 percent in the ring design. Sites that broke ground in 2019 or later show 33 and 35 percent. The ring design uses no near-miss control at all, so the gradient is not an artifact of the near-miss group being concentrated in recent years. It reflects where the projects were built. The earlier cohorts sit on much darker land, with a median pre-event radiance of 1.1 against 4.5 for the later cohorts, and a given amount of new lighting is a larger proportional change on a dark parcel. A cohort-robust estimator that compares each construction cohort only with never-built sites observed in the same calendar years, and therefore uses 17 treated sites with such support, gives 27 percent, with a 95 percent interval of $[-2, +92]$ percent. Its pre-construction leads reject jointly ($p = 0.008$), so we treat this estimator as inconclusive rather than as corroboration. The two better-supported designs above provide the evidence for the cohort pattern.

The choice of control group within the near-miss arm matters little. Restricting the controls to the nine isolated projects with a named hyperscale operator leaves the construction-year jump at 72 percent, against 70 percent in the full isolated sample, although that subset has a somewhat less flat pre-period ($p = 0.15$) and is too small to support estimates at longer horizons.

Table 5: The capital footprint across estimators and cohorts

Estimator	Construction year	Year 1	Built sites	Lead p
<i>Panel A: Built versus announced near miss, $\log(\text{NTL})$</i>				
All isolated sites	+70%	+231%	43	0.24
broke ground 2018 or earlier	+97%	+305%	26	0.41
broke ground 2019 or later	+33%	+140%	17	0.69
controls with a named hyperscale sponsor	+72%	+259%	43	0.15
Poisson on radiance levels	+59%	+153%	43	–
cohort-robust, never-treated controls	+27%	+154%	17	0.01
<i>Panel B: Within-site rings, $\log(\text{NTL})$</i>				
All isolated sites	+63%	+226%	43	0.51
broke ground 2018 or earlier	+95%	+294%	26	0.16
broke ground 2019 or later	+35%	+142%	17	0.89
<i>Panel C: Within-site rings, $\log(1 + \text{NTL})$, as reported in the text</i>				
All isolated sites	+38%	+132%	43	0.37
site-specific linear pre-trends	+38%	–	43	–
broke ground 2019 or later	+25%	+99%	17	0.90

Notes: Change in inner-ring (0–1 km) radiance at isolated hyperscale sites, in the construction year and one year later, transformed from log points to percent. Panels A and B use the same outcome, so their magnitudes are directly comparable; Panel C applies the $\log(1 + x)$ transform used in the text, which compresses changes at the low pre-construction radiance of these parcels. The within-site rows use no near-miss control at all, and they reproduce the cohort gradient in Panel A, so that gradient reflects differences between early and late construction cohorts rather than the thin near-miss coverage before 2020. Early cohorts sit on darker land, with a median pre-event radiance of 1.1 against 4.5 for later cohorts, which enlarges a log change. The cohort-robust row compares each construction cohort with never-built sites observed in the same calendar years, using the 17 sites with such support, and inference is a campus cluster bootstrap with 999 draws. Its construction-year 95 percent interval is $[-2, +92]$ percent, and its lead test rejects, so we treat that row as inconclusive rather than as corroboration. The site-specific-trend row fits a separate linear pre-trend to each site’s event years -4 through -1 and reports the observed-minus-predicted event-zero change; its campus-bootstrap 95 percent interval is $[+15, +71]$ percent. Lead p jointly tests event years -4 through -2 . Standard errors are clustered on two-kilometer campus groups throughout.

4.4 Spatial decay

The footprint is sharply localized. Estimated ring by ring, the effect is largest at 0 to 1 kilometer, falls by more than half at 1 to 2 kilometers, and is small and fading by 2 to 5 kilometers, with no detectable effect beyond five kilometers. A specification that omits the 0 to 1 kilometer ring and uses 1 to 2 kilometers as the inner comparison returns the same decay, so the result is not an artifact of the campus site itself. The footprint is large at the parcel and does not reach far.

4.5 Colocation facilities as a comparison

Colocation facilities in our sample are generally smaller, multi-tenant projects and often occupy or expand existing facilities, although some are purpose-built. They show no comparable change. On the construction-dated within-site design at 0 to 5 kilometers, the construction-phase estimate is -1.0 percent ($p = 0.65$) across 21 colocation facilities, against $+8.0$ percent ($p = 0.02$) across 43 hyperscale facilities, and the colocation operations-phase estimate is -1.4 percent ($p = 0.74$). The remainder of the paper therefore concentrates on hyperscale facilities and reports colocation as a comparison where the data support it.

5 Test of agglomeration

Section 4 establishes that a hyperscale data center is a large, spatially concentrated investment. The question is whether that capital generates the local spillovers that large plant openings have historically produced. [Greenstone et al. \[2010\]](#) trace those spillovers to three channels, namely a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. Each has been shown to raise productivity where firms cluster [[Combes et al., 2012](#)]. We test each channel at the facility level. Because the 25 to 50 kilometer control ring contains another data center at most sites (Section 3.3), the labor outcomes compare the pooled 0 to 25 kilometer zone with a 50 to 150 kilometer control around the same facility. Colocation facilities provide a second comparison, with the caveat that they sit in denser markets than hyperscale campuses. We find no detectable response through these channels. The salary and firm-entry estimates are precise enough to rule out modest effects; the hiring and supplier estimates are not.

Figure 2 puts the preferred estimates on a common scale. Advertised salary and firm entry are tightly bounded. Job postings are bounded more loosely, and supplier postings loosely enough that the estimate carries little information. Appendix Table 11 reports the corresponding intervals and minimum detectable effects.

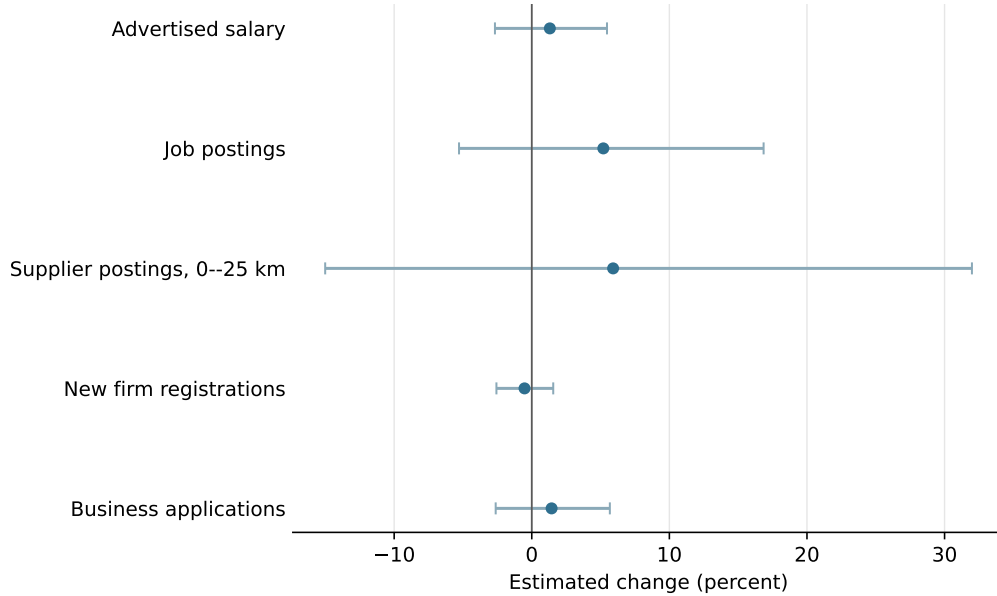


Figure 2: Precision of the agglomeration-channel estimates

Alt text: Horizontal confidence intervals for five estimates. Salary, firm-entry, and business-application intervals are narrow around zero; job-posting and supplier intervals are wider and include sizable positive effects.

Notes: Preferred estimate for each outcome with 95% confidence intervals, transformed from log points to percent. Salary, job postings, and supplier postings compare the pooled 0 to 25 km zone with a 50 to 150 km control. Firm registrations use the Virginia and Texas within-site trend break; business applications use the national county trend break.

5.1 Labor pooling

If a data center deepens the local labor pool, hiring and wages should rise in the surrounding area. Postings near built facilities move by +5.2% ($p = 0.35$) across 120 hyperscale facilities, with a 95 percent interval of $[-5.3, +16.8]$ percent. The built-versus-near-miss comparison agrees at +4.0% ($p = 0.75$) with a flat pre-trend ($p = 0.96$), as do Poisson counts, a long difference that controls for pre-period volume, and construction dating.

None of this is a precise zero, and it is worth being explicit about what the design can detect. The median hyperscale facility has about 760 advertised openings a year within five kilometers before it opens. An increase of 17 percent, the top of the interval, is roughly 130 additional postings a year. A hyperscale campus employs a few dozen people once operating, so a design of this precision cannot see the facility's own hiring, let alone distinguish it from a multiplier. The postings evidence rules out a large local hiring boom and is uninformative about anything smaller. The weaker within-site comparison was weaker still, admitting increases of roughly 30 percent.

Advertised salaries provide a much narrower test. The mean log salary of nearby postings moves by +1.3% ($p = 0.53$) across 119 facilities, with a 95 percent interval of $[-2.7, +5.5]$ percent. The panel is concentrated in salaried, posted positions, so the result is most relevant for white-collar workers and may not capture construction-trade compensation. The salary-disclosure rate is itself unchanged at opening ($p = 0.43$), so differential disclosure does not generate the result. Workplace

employment from the Longitudinal Employer-Household Dynamics data carries a different limitation. Its raw within-site effect is +8% at 0 to 5 km, but a strong pre-existing trend explains the increase, and detrending reduces the opening-date jump to -0.3% .

5.2 Supplier linkages

The second channel is local supplier linkages. We isolate postings in the sectors that build and equip a data center, namely construction and specialty trades, electrical-equipment wholesale, architectural and engineering services, and machinery manufacturing. Comparing the pooled 0 to 25 kilometer zone with the 50 to 150 kilometer control, the hyperscale supplier composite moves by $+5.9\%$ ($p = 0.61$), with a 95 percent interval of $[-15, +32]$ percent. Supplier postings are a thin series, and this interval is wide enough that the estimate does not establish a tight zero. It shows only that no large, precisely measured supplier cluster appears in the postings data.

5.3 Knowledge spillovers and firm entry

The third channel is knowledge spillovers among co-located firms. Appendix A maps heavy users of data-center capacity. They sit somewhat closer to data centers than the average firm, but much of this proximity reflects shared metropolitan location, and 69 percent of nearby compute firms pre-date the nearest facility. Exurban data centers often sit in places with few compute firms. These descriptive patterns do not identify an opening effect, but they give little indication of a cluster forming around the site.

New firm formation also shows no average jump at opening. We use secretary-of-state new-registration records for Virginia and Texas geocoded to the facility rings. The within-site jump at 0 to 5 km is $+0.7\%$ ($p = 0.24$) in the pooled sample and -0.5% ($p = 0.62$) for hyperscale facilities. A national county design using Census Business Formation Statistics also finds no effect ($+1.4\%$, $p = 0.49$). The upper bounds of the corresponding 95 percent intervals are 1.6 and 5.7 percent. We therefore find no systematic entry response. This does not imply that entry is zero in every location.

5.4 Incumbent productivity

In Greenstone et al. [2010], incumbent establishments experience total factor productivity gains of around 12 percent. We do not observe plant-level productivity. We instead ask whether advertised salaries or hiring at incumbent firms rise, two outcomes through which a large local productivity gain could appear. The advertised-salary interval has an upper bound of 5.5 percent, so it rules out a productivity gain of the size Greenstone et al. [2010] report only if more than about 45 percent of that gain reaches posted salaries. Pass-through below that leaves a 12 percent productivity gain inside our interval, so this outcome bounds the wage response rather than the productivity response itself. An event study of hiring by firms that existed before the facility shows no positive response, but the pooled and colocation samples carry a pre-existing downward trend that the design cannot

fully remove. We therefore read incumbent hiring as corroborating rather than causal evidence and do not claim to observe productivity directly.

5.5 What county-industry data can establish

County-industry data are a natural way to look beyond job postings, but the detailed Quarterly Census of Employment and Wages cells have two limitations in this setting. First, the Bureau of Labor Statistics suppresses employment when publication could reveal an individual employer. The public files record these cells as zero even though the employment level is not observed. Retaining the disclosure flag shows that suppression affects 68 percent of reported county-year records in data processing, 51 percent in telecommunications, and 66 percent in web and information services. Treating those values as observed zeros mechanically creates employment growth when a cell first becomes publishable.

Second, the host counties were already changing before the facility opened. We estimate cohort-by-event-time effects relative to the year before opening, using not-yet-treated counties as controls and a state-cluster bootstrap [Callaway and Sant’Anna, 2021]. Table 6 reports the average effect over event years zero through five and a joint test of the pre-period leads. Four of the ten hyperscale outcome series show statistically significant pre-opening changes at the five-percent level. These failures are concentrated in telecommunications, web and information services, and professional services. The data-processing establishment series is different: its pre-trend test does not reject, and the establishment count rises when the facility enters the county.

Table 6: County-industry disclosure and pre-trend analysis: hyperscale facilities

Industry (NAICS)	Employment suppressed	Employment effect	Pre-trend p	Establishments effect	Pre-trend p
Data processing (518)	68%	+5.2%	0.58	+27.2%***	0.11
Telecommunications (517)	51%	+5.2%	0.28	−2.6%	0.04
Web and information services (519)	66%	+9.3%	0.39	+18.3%***	0.03
Computer systems design (5415)	40%	−0.9%	0.85	+5.7%	0.11
Professional and technical services (541)	27%	+6.2%**	< 0.01	+7.6%***	< 0.01

Notes: Employment suppression is the share of reported county-sector-year records carrying the BLS non-disclosure code. Effects are cohort-weighted mean log changes over event years zero through five, transformed to percent, relative to not-yet-treated counties. Pre-trend p is a joint test for event years −4 through −2. Inference uses 1,999 state-cluster bootstrap draws. The 519 series ends in 2021 because the 2022 NAICS revision moved major components to sectors 513 and 516, so its post-period average is taken over a shorter window than the other sectors and is weighted toward earlier cohorts. Stars refer to the post effect: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Benjamini–Hochberg q -values across the ten sector-outcome tests are < 0.001 for data-processing establishments, 0.011 for web and information-services establishments, 0.066 for professional-services employment, and < 0.001 for professional-services establishments.

The county data identify one direct margin. Data-processing establishments rise by 27 percent, and the result survives a Benjamini–Hochberg adjustment across the ten sector-outcome tests.

This is the sector that contains the facility itself, so the increase confirms direct entry rather than downstream agglomeration. Data-processing and telecommunications employment do not show a precise increase once we preserve the suppression flags. The positive establishment estimates in web and information services and in professional services also survive the multiple-testing adjustment, but both have significant pre-period leads. We therefore do not use them as evidence of downstream growth. The county data distinguish direct entry from broader industry effects, but do not identify a downstream cluster.

6 Incidence outside the labor market

The housing evidence is suggestive rather than definitive. The within-site rent estimates are +6.6% at 0 to 5 km, +9.4% at 5 to 10 km, and +8.5% at 10 to 25 km. The gradient does not decline with distance, unlike the capital footprint, which is a first reason to doubt that these estimates isolate the facility. A density-matched placebo shows that comparable non-data-center locations also experienced substantial rent growth over the 2020 to 2023 work-from-home period, so the within-site estimates are upper bounds. Facility-level synthetic controls return smaller point estimates, between two and five percent across the three rings, although we cannot use this procedure for formal placebo inference. The built-versus-near-miss estimate is +4.3% ($p = 0.23$), with a flat pre-trend ($p = 0.44$). We therefore read the evidence as consistent with a modest rent response, with considerable uncertainty about its size. Colocation rent estimates show no detectable effect.

Neither housing supply nor net migration rises. A place-level Census Building Permits Survey analysis finds no multifamily permitting response, with permits for five or more units up 3.4% ($p = 0.87$). For migration, we assign each county its first reliably dated hyperscale opening within 25 km, including openings before the IRS panel, and estimate cohort-by-event-time effects using not-yet-treated counties. Gross inflows rise by 5.4% ($p = 0.02$) and gross outflows by 7.6% ($p < 0.01$), but the outflow series has a significant pre-trend ($p < 0.01$). In-migrant income also rises before opening ($p = 0.02$). We therefore do not interpret the gross-flow estimates as opening effects. The log inflow-to-outflow ratio for returns moves by -1.4% over event years zero through five ($p = 0.52$), with a joint pre-trend p -value of 0.14. The analogous adjusted-gross-income ratio moves by $+0.4\%$ ($p = 0.91$), with pre-trend $p = 0.09$. These estimates provide no evidence of net in-migration. The pre-opening trends prevent a causal interpretation of the gross migration or migrant-income estimates.

Foot traffic and commercial spending also show no robust change relative to near-miss sites. The foot-traffic estimate is about -12% ($p = 0.12$) on the full panel but attenuates to -3% ($p = 0.68$) on the balanced panel. Commercial spending moves by $+4.6\%$ ($p = 0.60$). The within-site spending estimate is positive and passes a random-donor comparison, but it is imprecise and the built-versus-near-miss design does not confirm it.

The fiscal estimates are concentrated in school finance rather than broad local-government spending. Table 7 reports the same not-yet-treated estimator and state-cluster bootstrap used in

the county analyses. Local school property-tax revenue rises by 16.6% ($p = 0.07$) with no detectable pre-trend. Capital outlay per pupil rises by 28.5%, but is imprecisely estimated ($p = 0.18$). Total local revenue and local revenue per pupil do not rise. The county highway estimate is positive and marginally significant, but moves materially when the opening dates are updated. The broad public-works series fails its pre-trend test. Neither the property-tax nor highway estimate survives adjustment across the six fiscal outcomes ($q = 0.21$ for each). We therefore treat the fiscal results as suggestive rather than as robust effects.

Table 7: Public-finance effects after hyperscale openings

Outcome	Geography	Effect	Post p	FDR q	Pre-trend p
Local property-tax revenue	school district	+16.6%	0.07	0.21	0.31
Total local revenue	school district	-2.3%	0.83	0.83	0.55
Local revenue per pupil	school district	+2.0%	0.76	0.83	0.14
Capital outlay per pupil	school district	+28.5%	0.18	0.35	0.57
Highway current + construction	county, balanced	+38.9%	0.07	0.21	0.15
Broad public works	county, balanced	+9.3%	0.40	0.61	< 0.01

Notes: Cohort-weighted mean log changes over event years zero through five, transformed to percent, relative to not-yet-treated geographies. School data are NCES/Census F-33 finance files for 2012–2023; county data are Census local government finance files for 2012–2023. Outcomes are adjusted for state-by-year means before the cohort comparisons. Inference uses 1,999 state-cluster bootstrap draws. Pre-trend p jointly tests event years -4 through -2 . FDR q applies the Benjamini–Hochberg adjustment across the six outcomes in the table. The highway estimate moves materially across alternative opening dates and is treated as a sensitivity result rather than a headline effect.

Taken together, the non-labor outcomes do not point to a broad local demand boom. The rent point estimates are positive and the fiscal point estimates are suggestive, but we find no net migration, permitting, foot-traffic, or spending response that would indicate a large expansion of the local economy.

7 Reconciliation with the concurrent literature

Three recent county-level studies bear directly on our result, and our evidence sharpens rather than overturns them. [Alvarez et al. \[2026\]](#) instrument data-center growth with a shift-share design and find positive effects on total employment, data-processing employment, construction, establishments, house prices, and electricity prices. [Yue and Zeng \[2026\]](#) report a similar pattern. Both attribute the gains to the facilities themselves, namely capital investment, construction, and specialized data-center employment, and neither claims agglomeration spillovers to unrelated firms. Our sub-county evidence agrees with that reading. The capital footprint is real and large, realized employment is confined to the data-center sector and its network, and we detect no spillover to surrounding firms. The papers reach the same conclusion through independent designs and at different spatial scales.

A second strand, including earlier work of our own using a synthetic-control design on county employment [Bahar and Wright, 2026], read a positive information-sector effect as evidence of local agglomeration in related information-technology activity. That reading does not survive finer data and a transparent estimator, and the revision is instructive. The information-sector gain decomposes into two parts. The larger part is data processing itself (NAICS 518), which is the data centers, together with telecommunications (NAICS 517), the network they carry. Both are direct rather than agglomeration. The remaining part is web and information services (NAICS 519), the one downstream component with a positive synthetic-control estimate. Three estimators agree that this component is a pre-trend rather than an effect. A within-county event study shows positive leads and a null jump, a staggered-robust design in the manner of Callaway and Sant’Anna [2021] returns an insignificant post-effect no larger than those leads, and the sector’s establishment count is flat before and after opening, so there is no firm entry. The synthetic control matched the pre-period level and read the continuation of a pre-existing trajectory as a treatment effect, a pattern its own pre-trend diagnostic did not flag.

The reconciliation is therefore not a contest between the prior papers and ours. It is an alignment with the instrumental-variables evidence that the local gains are direct, and a correction of the agglomeration interpretation that a synthetic control placed on a county-level information aggregate. The substantive conclusion, capital without agglomeration, holds across the within-site ring design, the realized county employment data, and a staggered-robust re-estimation of the one contested sector.

8 Interpretation

The results separate the physical scale of the investment from its propagation to nearby workers and firms. It is worth being explicit about what each half measures. The nighttime-lights result establishes that a large physical investment happened where and when we say it did, and that its visible extent stops within a few kilometers. It is a measure of the facility, including its security and parking lighting, substations, and cooling plant, rather than a measure of local economic activity. The labor, firm, and demand outcomes are the economic measures, and they are what the agglomeration question turns on. The contribution is the contrast between a capital footprint that is unambiguous and a set of economic responses that are not.

That contrast is uneven across outcomes, and the paper is only as strong as its tightest bound. Advertised salaries and new-firm formation show no opening-date increase and are bounded at 5.5 and 1.6 percent, so a response of the size the agglomeration literature reports would have been visible. Hiring is bounded at 17 percent, which excludes a large local boom but not a modest one. Supplier postings are a thin series and their interval is wide enough to be uninformative. The evidence therefore rules out a broad firm-entry or advertised-wage response more clearly than it rules out supplier gains, and we do not claim otherwise.

For a local multiplier of the conventional size to be hiding in these data, one of three things

would have to hold. The response would have to fall outside the zones we measure, which the 50 to 150 kilometer comparison is designed to catch. It would have to appear in employment that is neither advertised nor registered, which the county data would partly capture and do not. Or it would have to arrive later than our windows, which the post-2022 campuses will eventually reveal. What the technology suggests, and what the geography of compute in Appendix A supports, is that the channels themselves are weak here. Data centers and compute firms share metropolitan infrastructure and skilled labor, but 69 percent of compute firms within 25 kilometers predate the nearest facility, so the co-location is not by itself evidence that the facility created the cluster.

The non-labor evidence is more suggestive than decisive. Rent point estimates are positive in the synthetic comparisons, while net migration and multifamily permitting do not rise. School property-tax revenue and highway spending also have positive point estimates, but neither survives adjustment across the fiscal outcomes. Residential electricity rates do not rise for the cloud-era cohorts in our sample. These outcomes identify possible incidence margins, not a welfare accounting. We do not observe land prices, environmental damages, or the terms of utility and tax agreements.

Hyperscale and colocation facilities differ in both technology and location. Hyperscale campuses are generally larger and more likely to involve greenfield construction, while colocation facilities are multi-tenant and concentrated in denser markets. Their different outcome paths are informative, but they combine facility-type and market differences and should not be read as if facility type were randomly assigned.

9 Conclusion

A hyperscale data center rivals the manufacturing plants in the agglomeration literature in capital scale, but its visible local response is narrower. At isolated sites, brightness rises when construction begins and fades within five kilometers, and daytime imagery confirms that the physical change begins with construction. The announced near misses agree on the timing, although they are not balanced runner-up sites.

The labor and firm evidence separates direct entry from downstream clustering. Data-processing establishments rise with entry. Advertised salaries and new-firm registrations do not, with upper confidence bounds of 5.5 and 1.6 percent, and these bounds hold against a control area far enough away that the facility’s own cluster cannot reach it. Compute firms locate disproportionately close to data centers, yet most arrived first, and exurban campuses often have few such firms nearby.

A hyperscale project clearly brings large physical investment. For the cloud-era cohorts we study, we do not detect the nearby worker and firm responses that a broad local-multiplier claim requires, and the incidence we can see falls on land and public finance rather than on labor. Two limits are worth restating. Hiring and supplier postings are bounded loosely enough that moderate gains remain possible. And the post-2022 artificial-intelligence campuses, which are larger and more power-intensive than anything in our sample, will need their own evaluation. Whether they behave like the cloud-era build-out is an open question, and one that the subsidies now being written will

not wait for.

A The geography of compute

A common rationale behind data center incentives is that the facility will anchor a local digital economy: firms that use cloud capacity will locate nearby. We examine this claim by mapping where heavy users of compute locate relative to data centers and asking whether the shared location predates the facility. We use the Veridion firmographic database [Veridion, 2026], which geolocates 6.5 million US establishments and classifies each by industry, employment, and founding year. We define compute users by primary industry. Software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services form the compute-native core, with finance, scientific R&D, and semiconductors as a broader set.

A.1 Compute users cluster near data centers

Compute users are closer to data centers than the average firm. The median compute-native firm sits 11 kilometers from the nearest data center, against 20 kilometers for non-compute firms, and 69 percent of compute users fall within 25 kilometers of a data center, against 55 percent of other firms (Table 8). The association varies by industry. Cloud and hosting firms are nearest at a 6 kilometer median, software firms next at 9 kilometers, while telecommunications and streaming, whose footprints are more dispersed, sit farthest out.

This proximity is not an artifact of headquarters location. Measuring each firm’s operating footprint by the geography of its job postings rather than its headquarters returns the same pattern at a slightly larger distance, because a firm’s branch offices spread beyond its headquarters metro. Nor is it merely that both compute users and data centers crowd into a handful of tech metros. Removing the 45 largest metropolitan areas, compute users remain closer to the surviving data centers than non-compute firms are (a 75 versus 87 kilometer median, with software firms among the closest), so the association holds in second-tier and rural America as well.

Table 8: Distance from firms to the nearest data center

	Firms	Median km	≤ 25 km	≤ 50 km
<i>Panel A: Headquarters location (Veridion)</i>				
Non-compute firms	5,692,641	20.5	55%	70%
Compute users (core)	309,760	11.0	69%	81%
Cloud and hosting	26,188	6.1	75%	84%
Software	101,757	8.9	72%	84%
Computer systems design	154,934	11.7	68%	81%
Finance	193,540	15.9	62%	75%
Telecommunications	10,125	22.2	53%	66%
<i>Panel B: Establishment footprint (job postings)</i>				
Compute users (core)		15.1	63%	76%
<i>Panel C: Outside the 45 largest metros</i>				
Non-compute firms	2,603,195	86.8	20%	33%
Compute users (core)	103,356	75.3	30%	41%

Notes: Great-circle distance from each firm to the nearest reliably dated hyperscale data center. Panel A places each firm at its Veridion headquarters coordinate. Panel B replaces the headquarters with the firm’s job-posting locations (WageScape), weighted by the number of distinct hiring firms at each location, so it reflects the operating footprint rather than the headquarters. Panel C excludes all firms and data centers within 50 km of the 45 largest metropolitan areas and technology hubs and measures distance to the nearest remaining data center. Compute-native “core” sectors are software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services.

Compute users and data centers

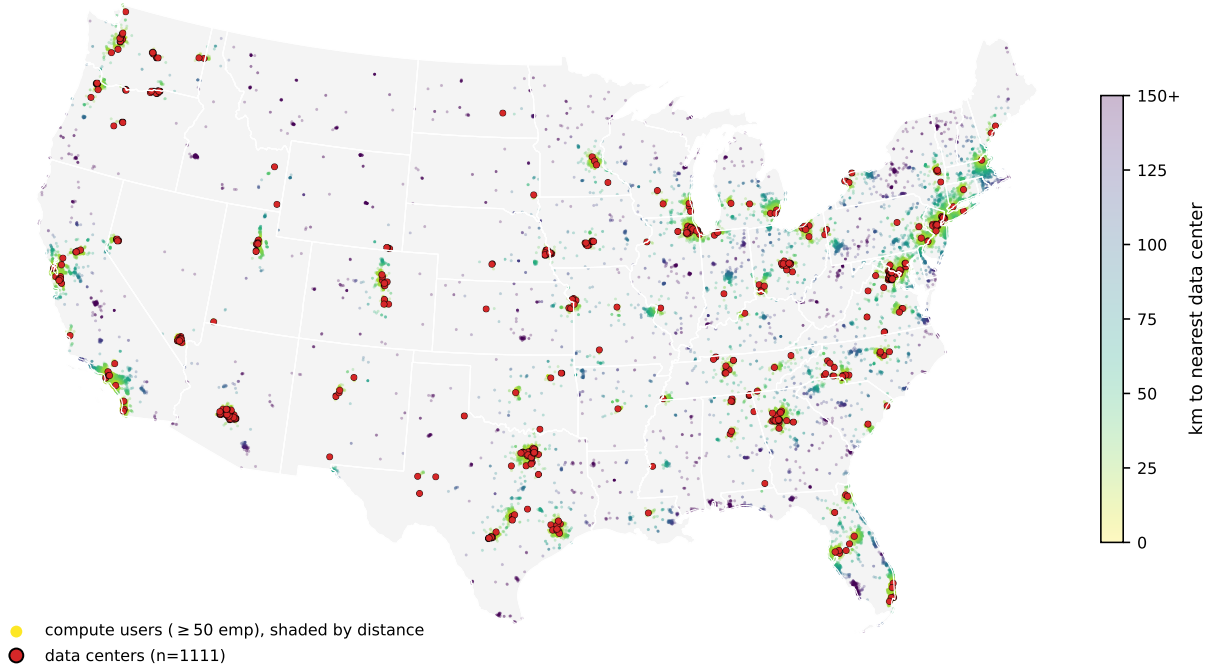


Figure 3: Compute-user firms and data centers across the continental United States. Points are compute-native firms with at least 50 employees, shaded by distance to the nearest data center, with data centers marked in red. Both concentrate in the same coastal and metropolitan corridors. Points in the interior are darker because they are farther from a facility.

Alt text: A US map shows compute firms and data centers concentrated in coastal and metropolitan corridors, with sparse coverage across much of the interior.

The two populations trace the same map (Figure 3), crowding into the coastal and metropolitan corridors and thinning together across the interior. The association is strongest for the largest firms. When sorted by employment, the median distance to a data center falls monotonically from 13 kilometers for the smallest firms to 4 kilometers for those with more than ten thousand employees, and the spread narrows with it. The biggest compute users cluster tightly around data centers, and almost none are far from one. Only about 5 percent of compute firms with at least a thousand employees sit more than 50 kilometers from any data center. The handful that do are software companies that scaled in place in second-tier cities, such as Esri in Redlands, Jack Henry in rural Missouri, and Ansys outside Pittsburgh. These examples show that large compute users need not locate near a data center to access cloud capacity.

A.2 Two kinds of data center hub

The geographic overlap is primarily a metropolitan phenomenon. Grouping data centers into hubs reveals a sharp split. Metropolitan hubs are embedded in dense clusters of sector-matched compute users. The 239 data centers around Ashburn, Virginia sit among thousands of government IT

contractors, with DXC, Leidos, Booz Allen, SAIC, and CACI all within 25 kilometers. The Bay Area hubs are ringed by Google, Broadcom, Synopsys, and ServiceNow. By contrast, the large exurban hyperscale clusters are isolated. The 50 data centers around Hillsboro, Oregon and the cluster near Quincy, Washington each have on the order of four compute firms within 25 kilometers. The same type of physical infrastructure coincides with a dense compute neighborhood in a technology center and a sparse one in the exurbs.

A.3 Shared location does not imply data-center-led clustering

Part of the explanation is shared infrastructure. Data centers and compute users both require access to the high-voltage grid, and both sit closer to it than ordinary firms. The median data center is 3.4 kilometers from a 230-kilovolt substation, the median compute user 5.0 kilometers, and the median other firm 6.4 kilometers. High-voltage substation density independently predicts where compute users locate. But the grid is only part of the story. Compute users remain closer to data centers after controlling for substation access. A shared pull toward the power backbone, fiber, and skilled labor can therefore place both in the same markets without one drawing the other.

Three patterns weigh against the interpretation that a data center creates the local compute cluster. First, the compute users near data centers predate them. Among compute firms within 25 kilometers of a data center, at least 69 percent were founded before the nearest data center opened. Because the firm registry over-represents recently founded firms, the share is likely higher. The local compute base usually existed first. Ashburn’s history is consistent with this ordering: Northern Virginia’s government technology firms and early internet exchange preceded the region’s data-center cluster. Second, the job-posting analysis finds no detectable increase around opening, either in compute-using industries or overall, although the estimates are imprecise. Third, exurban data centers are located in areas with few compute firms. The median metropolitan data center has roughly 2,700 compute firms within 25 kilometers, while the median exurban data center has 41 (Table 9).

Table 9: Compute users near metropolitan versus exurban data centers

	Data centers	Median compute firms ≤ 25 km	Share with < 5 nearby
Metropolitan data centers	824	2,702	0%
Exurban data centers	287	41	11%

Notes: Each data center is classified as metropolitan if it lies within 50 km of one of the 45 largest metropolitan or technology-hub centers, and exurban otherwise. The middle column reports the median count of compute-native firms (Veridion) within 25 km of the data center; the last column reports the share of data centers with fewer than five compute firms within 25 km. Separately, among compute firms within 25 km of any data center, 69 percent were founded before the nearest data center opened.

A.4 Implication for place-based policy

The evidence does not support the claim that a data center, especially an exurban one, will draw compute users to the surrounding area. Firms that consume cloud capacity locate near transmission, fiber, and skilled labor. These assets concentrate in metropolitan areas, and a single data center does not create them. The compute users already near data centers were mostly there first, and we find no systematic increase in firm entry after opening. The clearest measurable local response is the capital footprint, with more modest evidence of higher rents; we do not detect a cluster of digital firms.

B The instrument search

A natural way to address the siting-selection problem is an instrument for data center location. We searched for one and did not find a credible one. This appendix documents the search because the failure is informative. It is the reason the within-site design carries the main spatial interpretation, while announced near misses provide a benchmark rather than an instrument. The exercise also shows the limits of common siting predictors.

Predetermined siting instruments. We tested several plausibly exogenous, predetermined predictors of data-center siting, interacted where needed with the national capacity wave. We also applied a manufacturing falsification test: a valid instrument for data-center activity should not independently predict manufacturing employment. Table 10 summarizes the results. Transmission-substation density predicts siting strongly but fails this test because it is a proxy for industrial land. The 1980 urban college share, which works in the all-data-center county literature, carries the wrong sign for hyperscale, because hyperscale campuses go to low-college exurban land rather than to educated metros. Long-haul fiber proximity predicts siting only across metropolitan areas, not within a county, so it cannot identify a sub-county effect. Climate, measured by wet-bulb temperature and free-cooling days, does not predict siting at all, because the largest US hyperscale hub is hot and humid and operators engineer around climate to chase cheap power. Generation-capacity timing has no within-county predictive power. The features that predict hyperscale siting (power, industrial land, fiber) are the same features that are correlated with local industrial character. The features less likely to violate the exclusion restriction (climate and historical human capital) do not predict where these facilities go. We therefore do not find a credible predetermined instrument for hyperscale siting at the sub-county scale.

Table 10: Predetermined siting instruments and why each fails

Instrument	First stage	Why it fails
345 kV substation density	strong	predicts manufacturing employment
1980 urban college share	moderate	predicts fewer hyperscale sites
InterTubes fiber proximity	metro-scale	no within-county effect
Climate (wet-bulb, free-cooling)	none	does not predict siting
Generation-capacity timing	none	does not predict siting within counties

Notes: Each instrument is the predetermined county feature interacted with the national hyperscale capacity wave. Predicting manufacturing employment violates the exclusion restriction.

A shift-share instrument also fails. We also construct a shift-share instrument of the form used in concurrent county-level work [Alvarez et al., 2026], interacting predetermined fiber and college exposure with national data-center demand growth. This approach does not solve the problem because our public measures of data-center scale, namely capacity, square footage, and facility count, are lumpy and supply-driven. The first-stage F -statistic with square footage as the endogenous variable ranges from twenty to thirty, but the exclusion concerns described above remain.

Because the shift-share design does not resolve the exclusion concerns, we do not use its estimates as evidence. The instrument also predicts population-serving sectors such as health, reinforcing the concern that it captures general area development. We therefore report the exercise as documentation of the search rather than as an identification strategy.

C Incidence on electricity rates

A central concern in data-center siting debates is that a hyperscale facility’s load will raise electricity bills for nearby households. Existing estimates do not point in one direction. Wholesale-market modeling finds higher wholesale prices [Kay et al., 2026], utility-level estimates find higher retail rates, especially at investor-owned utilities [Meeks et al., 2026], and county-level instrumental-variable estimates find higher local electricity prices [Alvarez et al., 2026]. In contrast, Watten et al. [2026] estimate that data-center growth modestly reduced average retail rates from 2015 through 2024. Whether residential customers in a host utility’s territory pay more is therefore an empirical question. Data centers select utilities with cheap power and available capacity, so a comparison of treated and untreated utilities must account for the siting choice.

We treat each utility within each state as a separate unit. Using Energy Information Administration Form 861, we compute average residential prices as revenue over megawatt-hours sold from 2012 through 2024. A unit is treated in the year the first hyperscale facility opens in a county it serves. The hyperscale sample contains 52 utility-states first treated between 2015 and 2021, none of which also hosts a colocation facility. We compare them with roughly 2,000 utility-states that host no facility in the registry and, separately, with 54 utilities serving counties with an announced

hyperscale near miss. A separate colocation sample of 60 treated utility-states is reported alongside.

Figure 4 reports a cohort-by-event-time estimator using not-yet-treated controls. Against never-treated utilities, the mean effect over years zero through five is -1.0% ($p = 0.42$), with a joint pre-trend p -value of 0.97. The balanced panel, which retains utility-states observed in at least twelve of the thirteen years, gives -1.5% ($p = 0.17$). Against near-miss utilities, the full sample estimate is -0.9% ($p = 0.37$), and the year-five estimate is -1.9% ($p = 0.27$). Colocation estimates are small and positive, with a mean effect of $+0.7\%$ ($p = 0.54$).

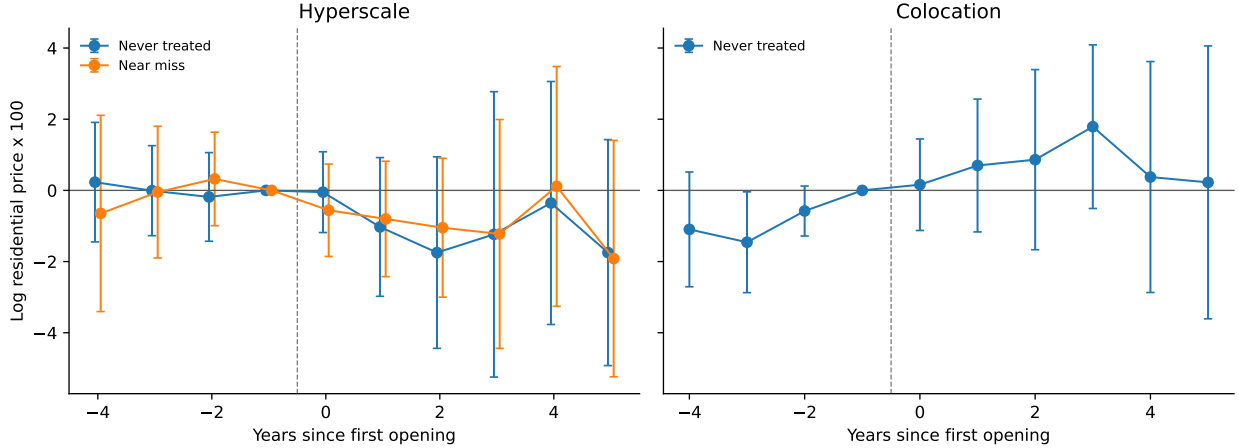


Figure 4: Residential electricity prices around the first data-center opening in a utility’s territory
Alt text: Two event-study panels show residential-price estimates around hyperscale and colocation openings. Hyperscale estimates become modestly negative after opening; colocation estimates fluctuate around zero with wide intervals.

Notes: Cohort-by-event-time estimates of log average residential price (revenue over MWh, EIA Form 861 bundled-service utilities, 2012–2024). Hyperscale: 52 treated utility-states, 2,040 never-treated controls, and 54 near-miss controls. Colocation: 60 treated utility-states and the never-treated controls. The never-treated comparisons remove state-by-year means; the near-miss comparison removes year means. Inference uses 1,999 state-cluster bootstrap draws. Reference period is the year before opening; 95% confidence intervals shown.

The estimates provide no evidence that host-utility residential rates rise over this horizon. The negative point estimates are consistent with fixed network costs being spread over a large new load, but the full-sample and near-miss averages are imprecise. We do not observe the utility’s cost allocation or contract, so the design does not identify that mechanism. Retail rates are also only one incidence margin: electricity generation creates environmental costs that may be borne outside the host utility [Muller, 2026]. The results cover cloud-era cohorts through 2021 followed to 2024, not the larger post-2022 artificial-intelligence build-out.

D Supplementary results

Table 11 reports confidence intervals and minimum detectable effects for the preferred agglomeration estimates. Table 3 in Section 3.3 reports the same outcomes under the conventional within-site

control ring, and Table 5 reports the capital footprint across estimators, transforms, and construction cohorts.

Table 11: Precision of the agglomeration-channel estimates

Outcome	Comparison	Units	Estimate	95% CI	80% MDE
Advertised salary	0–25 vs 50–150 km	119	+1.3%	[-2.7, +5.5]	5.9%
Job postings	0–25 vs 50–150 km	120	+5.2%	[-5.3, +16.8]	16.2%
Supplier postings, 0–25 km	0–25 vs 50–150 km	120	+5.9%	[-15.0, +32.0]	37.0%
New firm registrations	Within-site trend break	117	-0.5%	[-2.6, +1.6]	3.0%
Business applications	National county	38	+1.4%	[-2.6, +5.7]	6.0%

Notes: Preferred estimate for each outcome, transformed from log points to percent. MDE is the minimum two-sided effect detectable with 80 percent power at the five-percent level, holding the reported standard error fixed. Salary, job postings, and supplier postings compare the pooled 0–25 km zone with a 50–150 km control, because another data center sits inside the 25–50 km ring at 88 percent of built sites (Table 3). Firm registrations use the Virginia and Texas within-site trend break; business applications use the national county trend break. The unit count is facilities for the ring designs and treated counties for the business-application design.

Data Availability Statement

This study combines publicly available data with several proprietary datasets obtained under license. All code that constructs the analysis datasets from the raw sources and reproduces every table and figure will be deposited in a public data and code repository. The deposit includes the publicly available inputs and all author-constructed files, namely the data-center registry, the operating and construction-start dates, and the announced near-miss cohort. The proprietary inputs cannot be redistributed. For each, we report the provider and access terms and include the exact product identifiers, date ranges, geographic filters, and extraction code, so that a licensed user can regenerate the facility-by-ring panels and reproduce all results.

Publicly available data.

- Satellite nighttime lights: VIIRS Day/Night Band Monthly Composite (NOAA, product VCMCFG), accessed through Google Earth Engine.
- Daytime imagery for construction dating: Sentinel-2 and Landsat surface reflectance (ESA and USGS), accessed through Google Earth Engine.
- Data-center registry: starting point from the IM3 Open Source Data Center Atlas (Pacific Northwest National Laboratory), augmented by the authors from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, trade press, DatacenterMap, Cloudscene, and data-center REIT filings. The compiled registry, dates, and announced near-miss cohort are in the deposit.
- Construction timing: Federal Aviation Administration Obstruction Evaluation / Airport Airspace Analysis filings (Form 7460-1).

- Employment and establishments: Quarterly Census of Employment and Wages (Bureau of Labor Statistics) and County Business Patterns (Census Bureau).
- Migration and income: IRS Statistics of Income county-to-county migration data.
- Housing supply: Census Building Permits Survey. Population: 2020 Census Centers of Population (tract population-weighted centroids).
- Workplace employment: LEHD Origin-Destination Employment Statistics (LODES, Census Bureau).
- Business formation: Census Business Formation Statistics, and state business-registration bulk files for Virginia, Texas, and Iowa (Secretaries of State and data.iowa.gov).
- Local public finance: Census of Governments and the Annual Survey of School System Finances (F-33).
- Electricity: U.S. Energy Information Administration Form 861.
- Instrument-search inputs: the InterTubes long-haul fiber map [Durairajan et al., 2015]; the 1980 county urban-college share (IPUMS NHGIS); and transmission-substation locations (Homeland Infrastructure Foundation-Level Data).

Proprietary data (available under license, not redistributable).

- Job postings: LinkUp Job Records [LinkUp, 2025]; WageScape salary postings [WageScape, 2022].
- Residential rents: RentHub Rental Data [RentHub, 2022].
- Consumer spending: SafeGraph Spend Patterns [SafeGraph, 2022].
- Foot traffic: Advan Weekly Patterns [Advan Research, 2025].
- Firmographics: Veridion firm master data.

The five Dewey-distributed datasets are available to researchers through a subscription to Dewey Data (<https://www.deweydata.io>); Veridion firmographics are available under license from Veridion (<https://veridion.com>). The authors accessed these data under institutional license and are not permitted to redistribute the raw files. The deposit provides the dataset identifiers and the extraction and aggregation code required to rebuild the analysis files from a licensed copy.

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